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Hub location problems: A meta review and ten disruptive research challenges

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As global air transportation as well as supply chains face unprecedented challenges, optimizing hub locations is crucial for enhancing efficiency, reducing costs, and mitigating environmental impacts. This paper reviews the state-of-the-art in hub location problems, with a particular focus on air transportation, driven by the increasing complexity and importance of networks in today's interconnected world. Our review offers two major contributions: a meta-review of existing surveys to synthesize current knowledge and identify gaps, and the proposal of ten critical research challenges encapsulated in the acronym DISRUPTIVE. These challenges include the need for high-quality datasets, deeper analytical insights, sustainability considerations, robustness against disruptions, addressing uncertainty, leveraging parallelization techniques, incorporating temporal dynamics, embracing interdisciplinary approaches, designing versatile hubs, and exploring emerging transportation modes. By addressing these challenges, researchers can drive innovation in hub location problems, paving the way for more resilient and efficient logistical networks that meet the demands of a rapidly evolving global landscape.

Contents

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1. Introduction

Hub location problems (HLPs) aim to optimize the location of hub facilities in a network as introduced by O'Kelly (1986). These hub facilities collect flows between origins and destinations, while, by switching/sorting/collecting and a consolidation/breakbulk function, exploiting economies of scale (Campbell & O'Kelly, 2012). The strategic placement of hubs and the allocation of demand flows are crucial challenges in network design, often addressed through mathematical optimization techniques. Research on hub location problems has received considerable contributions from interdisciplinary researchers, given their applicability in many different domains, mainly transportation (e.g., Gelareh et al., 2015; Gelareh & Nickel, 2011; O'Kelly, 2012) and telecommunication (e.g., Kim & O'Kelly, 2009; Yaman & Carello, 2005). The widespread interest over multiple decades has led to a plethora of research papers. Early works by O'Kelly (1987) and Campbell (1994a, 1994b) established the foundation for discrete HLPs, drawing inspiration from classical p-median and facility location models (ReVelle & Swain, 1970). These models have been adapted to air transport, where hubs serve as central sorting points for passenger flows and cargo shipments (Bryan, 1998). Two critical considerations in hub location models are single-assignment vs. multiple-assignment structures (Campbell, 1992), or its blend of r-allocation (Yaman, 2011). In a single-assignment model, each origin–destination (OD) pair is routed through a single hub, which simplifies network control but may lead to inefficiencies. Conversely, multiple-assignment models allow dynamic allocation of flows to multiple hubs, improving flexibility but increasing computational complexity (Contreras et al., 2011). While this assumption was attractive for initial modeling purposes, it is much more likely that aviation networks serve passengers who desire multiple routing options through accessible hubs. It is clear however, that air freight can take advantage of single allocation in some cases. In the comparatively difficult case of single assignment, theoretical advancements in quadratic assignment problems (QAPs) have influenced hub location optimization (O'Kelly, 1992a). The non-convex nature of quadratic hub models poses computational challenges, necessitating specialized algorithms (Hansen, 1972). Additionally, elastic demand models have emerged as a key research area, incorporating dynamic passenger choices based on ticket prices, convenience, and airline competition (Taner & Kara, 2016).

In this study, we review the state of the art in hub location problems. Existing reviews on hub location problems often suffer from fragmentation and limited scope. They are frequently published across diverse venues, making it challenging to obtain a comprehensive overview. Moreover, these reviews predominantly focus on the dissection of primary research studies, overlooking the synthesis and critical evaluation of existing surveys; surveys which are often cited/acknowledged but not built upon. This state of the literature prevents a lack of cohesive insights and hinders the identification of overarching trends and gaps in the field, especially for newcomers. Finally, existing reviews put a strong backward-looking emphasis, describing past formulations and trends, without a rich discussion of future work. Here, we provide a more integrated and reflective review to advance the understanding and development of hub location problems. We have limited the look-back towards primary existing studies to the minimum, only briefly reviewing existing models and solutions techniques, without going into details at the formulation/algorithm level, given that existing reviews did an excellent job in this regard. Instead, we put a much stronger focus on the aspects of a meta-review across existing recent surveys, together with the derivation and discussion of ten disruptive research challenges. Our review and challenge descriptions are intentionally chosen to provide a comprehensive foundation that underscores the broader applicability and significance of hub location problems, which are inherently interdisciplinary. By adopting such a more general approach, our review not only caters to the air transport community but also invites insights from other fields, thereby enriching the discussion and potential solutions. This broader perspective is crucial for highlighting the universal challenges and innovative methodologies that can be adapted and specialized for air transportation. Furthermore, it ensures that the paper remains accessible and relevant to a wider audience, fostering cross-disciplinary collaboration and advancing the state-of-the-art in hub location research. Overall, we believe that our work makes a seminal contribution to the literature, by a clear dissection of the state of the art as well as open research questions.

The remainder of this study is structured as follows. Section 2 presents a brief overview regarding hub location problems, covering model types and solution techniques. Section 3 discusses recent surveys/reviews on HLPs, providing a meta-review of subjects and suggestions for future research directions covered in the extant literature. Section 4 introduces ten challenges we believe are important for future research in this area. Section 5 concludes this study.

2. Overview on HLPs

Hub location problems are typically categorized by their structures and objectives, addressing various challenges in network design. Research in this area has evolved through different optimization models, each focusing on specific cost and efficiency trade-offs. A foundational model is the p-hub median problem (pHMP), which assumes a fixed number of hubs (p) and aims to minimize transportation costs by optimizing flow routing (Campbell, 1996; Ernst & Krishnamoorthy, 1996). Studies by De Camargo et al. (2009), Lin and Lee (2010), Matos (2021) have refined this approach, with application to airline networks to enhance operational efficiency. In contrast, the Hub Location Problem with Fixed Costs (HLP) offers more flexibility by determining both the number and placement of hubs, considering both transportation and setup costs (O'Kelly, 1992b). This model, explored by Stanimirović (2007) and Abyazi-Sani and Ghanbari (2016), is widely used in airline network design, especially where demand patterns vary. Since the number of hubs is not predetermined in HLP models, additional constraints like capacitated hubs are often introduced to provide structure. Further advancements in hub location research involve the structure of inter-hub connections. Traditional models often feature fully interconnected hub networks, where all hubs are linked. However, studies by O'Kelly et al. (2015) and de Camargo et al. (2017) have explored incomplete hub networks, where only a subset of hubs are connected, reducing costs and increasing flexibility. These incomplete networks can be designed through exogenous approaches, using predefined structures like ring or hierarchical models, or endogenous approaches, where inter-hub connections emerge dynamically during optimization. Fig. 1 visualizes a toy example for a network instance with five hubs, multiple allocation, an incomplete hub network and a few direct links bypassing hub traffic.

Table 1 reviews various solution techniques for hub location problems, each suited to different input sizes and problem complexities. Exact methods, such as Integer Programming (IP), provide high-accuracy solutions but are computationally expensive for large-scale problems, making them ideal for small to medium-sized hub networks where precision is critical. Benders Decomposition is efficient for large-scale problems with decomposable structures, particularly for uncapacitated multiple allocation hub location problems, where the problem can be divided into more manageable parts (Contreras et al., 2011; de Camargo & Miranda, 2012; de Camargo et al., 2009, 2008; Rodríguez-Martín & Salazar-González, 2008). Branch-and-Bound (B&B) guarantees optimality but is slow for large instances, suitable for solving single allocation p-hub median problems where exhaustive search is feasible. Branch-and-Cut (B&C) improves on B&B by adding valid inequalities, though it remains computationally intense, and is used for large-scale problems with additional constraints (Labbé et al., 2005; Rodríguez-Martín & Salazar-González, 2008; Rodríguez-Martín & Yaman, 2014). Lagrangian Relaxation reduces complexity by providing lower-bound

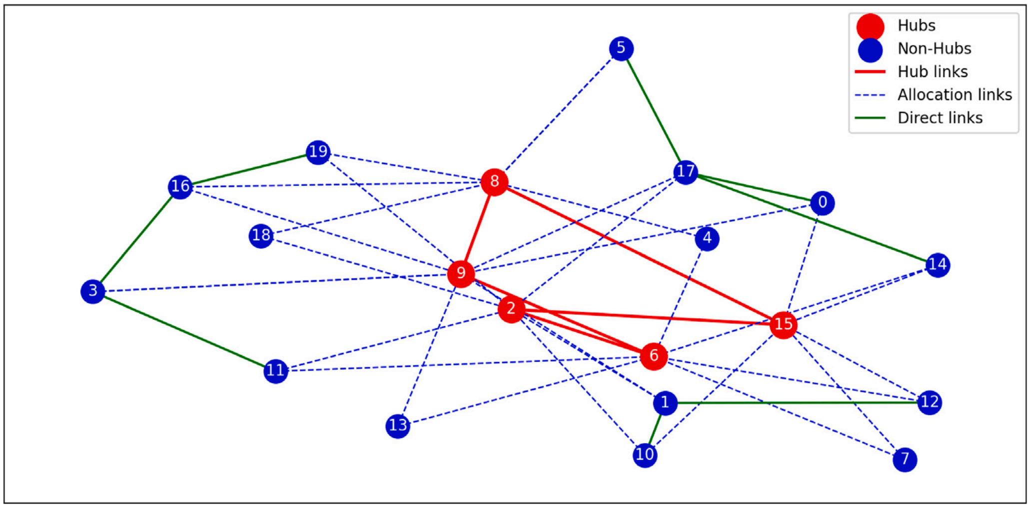

Fig. 1. Synthetic example of a hub network with five hubs (red circles), 15 spokes (blue circles), five hub links (red lines), on average two allocation links (blue lines), and various direct links (green lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Summary of solution techniques for hub location problems.

Solution Technique	Type	Computational Trade-offs	Typical Applications
Integer Programming (IP)	Exact	High accuracy but computationally expensive for large-scale problems	Used for small to medium-sized hub networks
Benders Decomposition (Meraklı& Yaman, 2016; de Sá et al., 2018a, 2018b)	Exact	Efficient for large-scale problems with decomposable structures	Solving uncapacitated multiple allocation hub location problems
Branch-and-Bound (B&B) (Ernst & Krishnamoorthy, 1998; Gendron et al., 2016)	Exact	Guarantees optimality but slow for large instances	Solving single allocation p-hub median problems
Branch-and-Cut (B&C) (Labbé et al., 2005; Rodríguez-Martín & Salazar-González, 2008)	Exact	Improves B&B by adding valid inequalities, still computationally intense	Large-scale hub location problems with additional constraints
Lagrangian Relaxation (Rostami et al., 2016; Tanash et al., 2017)	Exact	Reduces complexity but provides lower-bound solutions	Used in capacitated hub location problems
Genetic Algorithms (GA) (Kratica et al., 2011, 2005; Stanimirović, 2007)	Heuristic	Good for large problems but does not guarantee optimality	Finding near-optimal solutions in multi-hub networks
Tabu Search (TS) (Abyazi-Sani & Ghanbari, 2016; Karimi, 2018; Skorin-Kapov & Skorin-Kapov, 1994)	Heuristic	Avoids local optima, but performance depends on fine-tuning	Applied in airline hub network optimization
GRASP (Peiró et al., 2014)	Heuristic	Fast convergence but solution quality depends on randomness control	Logistics and supply chain network design
Variable Neighborhood Search (VNS) (Brimberg et al., 2017; Dai et al., 2019; Ilić et al., 2010; Soylu & Katip, 2019)	Heuristic	Good for large problems but does not guarantee optimality	Applied in airline hub network optimization

solutions, useful in capacitated hub location problems where approximate solutions are acceptable (Contreras et al., 2009; Elhedhli & Wu, 2010; Fisher, 2004). Heuristic methods offer flexibility for larger problems. Genetic Algorithms (GA) are effective for large-scale problems, though they do not guarantee optimality, making them suitable for finding near-optimal solutions in multi-hub networks (Azizi et al., 2016; Cunha & Silva, 2007; Kratica et al., 2005; Stanimirović, 2012). Tabu Search (TS) avoids local optima but requires fine-tuning, commonly applied in airline hub network optimization where adaptability is key (Abyazi-Sani & Ghanbari, 2016; Calık et al., 2009; Skorin-Kapov & Skorin-Kapov, 1994). Variable Neighborhood Search (VNS) is also effective for large problems, balancing exploration and exploitation, and is applied in airline hub network optimization (Ilić et al., 2010; Todosijević et al., 2017). The choice of solution technique depends on the problem size, required precision, and computational resources.

3. Meta-review of extant HLP survey/review papers

Numerous prior reviews have been published on the subject; reviews since the year 2008 are shown in Table 2. Given that these reviews were published in distinct communities/journals, it is possible that they contain overlapping information and insights. Moreover, they have a strong focus on summarizing primary studies, without a broad embedding into existing reviews. Therefore, we provide a summary of each review's focus and major suggestions for future research here. This is - to the best of our knowledge - the first comprehensive meta review in the field of hub location problems.

Alumur and Kara (2008) contributed a seminal survey that is concerned with the classification and overview of network hub location models. In these models – contrary to continuous hub location problems on a plane – deals with n nodes on which the set of origins, destinations

Table 2
Summary of prior seminal reviews on HLPs published since 2008.

Authors	Emphasis
Alumur and Kara (2008)	Classify discrete HLP model variants
Kara and Taner (2011)	Outlines the research on HLPs
Campbell and O'Kelly (2012)	Historical assessment of 25 years anniversary
Farahani et al. (2013)	Reviews recent advances in HLP
Essaadi et al. (2016)	Analytical grid along nine hub location dimensions
Contreras and O'Kelly (2020)	Hub Location Problems
Alumur et al. (2021)	Framework for state of the art and outlook
Khaleghi and Eydi (2022)	Multi-period hub location problems
Wandelt et al. (2022)	Benchmark of formulations and techniques
Sharma et al. (2025)	Bibliometric study and citation analysis

and potential hub locations are identified. A large part of the review is devoted to a detailed discussion of formulations, their components and how their use deviates across the surveyed papers. Particularly, the authors cover allocation types, setup cost-based problems, p-hub center, hub covering problems, and the role of the discount factor. Overall, over 100 papers are reviewed and categorized. The authors conclude with a synthesis and discussion of future research directions, which cover mainly gaps in specific combinations of formulation components, the use of multiple, possibly conflicting objectives, and the need to develop more realistic studies that address real-life situations.

Kara and Taner (2011) provides a discussion of the historical development of HLPs, starting with the importance of consolidation and dissemination points as well as their endogenous effects in spatial interaction theory, explaining the seminal contribution of O'Kelly (1986) in greater detail. Based on this reference, the authors proceed to a propose new taxonomy that serves for a convenient classification of the relevant developments, covering cost minimization and minmax type of objectives. Moreover, the authors include a complementary discussion on more application-oriented literature, concluding with a set of recommendations for current trends and future work. These cover a consideration of competition as well as the incorporation of a non-linear cost function.

Campbell and O'Kelly (2012) celebrated the 25th anniversary of two seminal transportation hub location publications, which appeared in the year 1986. The authors explicitly state that their work is to reflect on the origins of hub location research, especially in transportation, and provide some commentary on the present status of the field. This study surveyed the extant literature, covering formulations and applications of fundamental hub location problems as well as the motivations and inherent methodological issues (e.g., flow-independent discounts or complete hub networks). The current research endeavors are summarized into the categories of solving large scale problems, alternate hub/allocation topologies, the integration of costs/service, dynamic/competitive hub location problems, and models with stochastic elements. Concerning future research directions, the authors motivate the use of more realistic transportation costs and service measures. Moreover, the highlight potential interdisciplinary aspects and different habits in different research communities, particularly regarding the presentation and documentation of results.

Farahani et al. (2013) focused on reviewing the most recent advances in HLP from between the year 2007 and 2013, building upon the survey of Alumur and Kara (2008), which was focused on network-type hub location models. A large part of the review, that covers over 160 papers, revisits mathematical model formulations as well as applications in real-life case studies. The authors provide a highly detailed overview on the used solution techniques in each paper, distinguishing exact methods and heuristics; and also information about the scale of the tested instances. Moreover, the authors identify various gaps in the literature, covering aspects of downstream traffic generation (hubbing effect), the possibility of skewed flows towards non-hub connections, the potential for plane-based formulations, the incorporation of multi-criteria decision making, as well as domain-specific model extensions. Finally, the authors discuss a few future research areas that cover reliability, sustainability, global logistics applications, the combination of HLP with other location problems, uncertainty, as well as the absence of methods that scale up to very large instances, together with the need for effective bounding techniques.

Essaadi et al. (2016) conducted a comprehensive literature review that highlights the diverse approaches and underlying assumptions prevalent in existing research. The authors propose a mapping of these aspects to create a general analytical framework, or "grid", for organizing the literature. This grid is based on the premise that the relative positioning of research papers facilitates a better understanding of their commonalities and differences, while also emphasizing the often-overlooked contingencies, as authors infrequently explicate all their assumptions. The developed grid is structured into nine dimensions: scope of the paper, decision-making framework, hub features, logistics network structure, transport organization, physical data, economic evaluation system, decision evaluation criteria, and application field. This survey not only categorizes existing research but also identifies potential novel research themes along the grid's dimensions.

Contreras and O'Kelly (2020) systematically summarized the key features, assumptions, and properties of HLPs, underscoring the critical importance of location and network design decisions. The authors review recent trends and advancements in hub location research, encompassing network topologies, cost structures, capacitated models, uncertainty, dynamic and multi-modal models, and competitive and collaborative strategies. Notably, they highlight successful integer programming formulations and efficient algorithms for solving HLPs. The review emphasizes the need for additional models to better represent the optimal evolution of hub networks and transportation mode choices, as hub networks are typically not redesigned from the scratch. Given the significance of competition and collaboration in most hub location applications, the authors suggest further investigation into models that consider competitive environments, collaborations, mergers, and acquisitions.

Alumur et al. (2021) provided an overview on the state of the art in hub location research, through taxonomy of problem types, while identifying key gaps that provide opportunities for better models. The authors have a focus on a historical perspective, the development of economies of scale, as well as application areas and how they have been extended over time. Moreover, they review the hub location formulations and solution techniques. Concerning perspectives for future research, the authors highlight recent success in the development of solving realistic problem sizes but acknowledge that many models still incorporate various simplifying assumptions or idealizations that significantly reduce the extent of realism from the real networks being modeled. Key themes for future research, according to the authors, include a better modeling of economies of scale, the incorporation of time, more sophisticated objectives/multiple criteria, more realistic models that reflect either structural changes in the industry or significant changes in travel patterns, the integration of HLP with other problems, incorporating the nature of demand, the use of more real /realistic data, transforming results to insights, and the need to understand which formulations and methods work best in which case.

Khaleghi and Eydi (2022) developed a comprehensive review of multi-period hub location problems from 1990 to the year 2022. In contrast to the static hub location problem, where the decisions are made once for all periods, decisions in the multi-period hub location problem are made according to the changes in parameters in every period. The authors' framework for discussion is based on the following characteristics: type of planning horizon (continuous-time and discrete-time), capacity constraints, type of problem (median and covering), type of services (single-level and hierarchical), number of commodities and modes, type of hub facility (mobile and virtual), type of assignment, and type of parameters (deterministic and uncertain). Besides a formal comparison of model formulations and solution techniques, the authors also discuss real-world applications Finally, a set of future research topics is proposed, covering multi-objective models, sustainability aspects, a stronger focus on continuous-time models, the incorporation of uncertainty/waiting times at hubs, and the consideration of competitive/pricing models.

Wandelt et al. (2022) presented a reference experimental benchmark for solving HLPs, which are inherently NP-hard due to their combination of facility location and quadratic assignment problem aspects. The study reports on an extensive benchmarking and reproduction effort, investigating twelve fundamental HLP variants that incorporate single or multiple allocations, p-hub median objectives or fixed hub setup costs, capacitated or uncapacitated hubs, and complete or incomplete networks. The authors implemented four standard exact algorithms – including CPLEX, three variants of Benders decomposition, and row generation – and evaluated them on subsets of three standard datasets (CAB, TR, and AP), yielding over 5,000 optimal solutions. The computational analysis encompasses scalability, convergence, iteration analysis, runtime deviation drivers, memory usage, optimal solution patterns, and hub flow patterns. This work identifies several avenues for future research, including the selection of appropriate solution methods, improved prediction and explanation of extreme computation times, a focus on scalability, and the integration of data mining and artificial intelligence techniques.

Sharma et al. (2025) conducted a comprehensive bibliometric literature review to develop a thematic framework that describes and presents hub location problems. Major themes covered in their review include cooperation, coopetition, sustainability, reshoring, and dynamic demand, which are said to contribute to the complex challenges in today's hub location problems. The authors put a strong emphasis on the evolution of hub location research, prominent research clusters, influential authors, leading countries, as well as employed keywords. In conclusion, the review offers a set of future research directions /challenges, covering the aspects of formulation/solution approaches (integration of machine learning and dynamic factors), the trade-off between optimal solutions/heuristics, the development of sustainable hub location models, the incorporation of collaborative/coopetitive hub networks and models for their formation and coordination, as well as the impact of emerging technologies in location decision making (e.g., drones, e-commerce, shared mobility, and data analytics).

Table 3 provides a summary of the future research proposed by the major surveys. It can be seen that some of these directions reappear, e.g., the inclusion of competitive aspects, incorporation of uncertainty, and also scalability.

4. Challenges towards DISRUPTIVE research on HLPs

In this section, we provide ten suggestions for future disruptive research on hub location problems. These suggestions were derived based on the aggregated recommendations in extant survey and the research experiences of the authors of this study. The dissemination of additional datasets (Section 4.1) and insight-based research (Section 4.2) enhance the accuracy and holistic nature of hub location models, reflecting real-world dynamics and integrating diverse insights. Sustainability (Section 4.3), robustness (Section 4.4), and the incorporation of uncertainty (Section 4.5) are crucial for developing resilient hub networks that minimize environmental impact and adapt to disruptions as well as dynamic environments, supported by parallelization techniques that improve computational efficiency (Section 4.6). Incorporating temporal aspects (Section 4.7) and interdisciplinary approaches (Section 4.8), such as those using machine learning and AI, allows hubs to respond dynamically to demand fluctuations and optimize operations. Versatile hub designs (Section 4.9) and the integration of emerging transportation modes (Section 4.10), like drones and autonomous vehicles, increase operational flexibility and present new challenges, ultimately enhancing the overall efficiency and resilience of logistics networks. Overall, the first characters of the ten words "Datasets", "Insights", "Sustainability", "Robustness", "Uncertainty", "Parallelization", "Temporal aspects", "Interdisciplinary aspects" "Versatile hubs", and "Emerging modes" collectively spell out the word DISRUPTIVE. Below, we describe the ten research suggestions in greater detail.

4.1. Datasets

The availability of comprehensive datasets is essential for advancing HLP research (Wandelt et al., 2022). Existing research has mostly relied on 3-4 established datasets, namely AP, CAB, and TR. These datasets have limitations in terms of scale and available additional data. Modern logistics and transportation networks generate vast amounts of data, which can be leveraged to create more accurate and dynamic models for future research. By integrating richer data on traffic patterns, demand fluctuations, and operational metrics, researchers can use these datasets to develop predictive models that enhance decision-making processes. Moreover, access to diverse datasets enables the validation of theoretical models against real-world scenarios, ensuring that solutions are practical and robust. The dissemination of Open Data regarding HLP research is a highly promising avenue for future research which may support the development of more expressive and realistic models. In fact, much of the state of the art in HLPs would not be possible if it was not for the release and re-use of AP, CAB, and TR in the HLP community. We explain the opportunities for future research and its challenges in greater below.

Traditional datasets often rely on artificial setup costs, which, despite various formulae in the literature, fail to capture the nuanced financial dynamics of establishing hubs in diverse geographical and economic contexts. By incorporating real-world cost data, researchers could be able to develop models that more accurately reflect the financial implications of hub placement, leading to more practical and implementable solutions. Similarly, capacity generation in current datasets is often artificial, with tight and loose versions that do not clearly translate into real-world scenarios, but seem to be more motivated by generating hard and easier instances. This artificiality can lead to models that either overestimate or underestimate the capacity requirements, resulting in inefficient resource allocation. Richer datasets that include real-world capacity data can help bridge this gap, enabling models to better align with actual operational constraints and demands. Incorporating more accurate distance metrics that account for factors such as terrain, traffic, and infrastructure can significantly enhance the practical applicability of hub location models. This is particularly relevant in intermodal and multi-modal application scenarios, where the integration of different transportation modes (e.g., road, rail, air) requires a more nuanced understanding of distance and cost trade-offs. Temporal data is another critical aspect that is often overlooked in traditional datasets. Hub location problems are dynamic, with demands and costs changing over time. Incorporating temporal data can help capture these dynamics, enabling models to adapt to seasonal variations, economic cycles, and other time-dependent factors. This can lead to more robust and resilient hub location strategies that can withstand temporal fluctuations. Finally, the uncertainty of parameters in hub location problems is a significant challenge that needs to be addressed. Richer datasets that incorporate probabilistic or stochastic elements can help model this uncertainty, enabling researchers to develop more robust and adaptable solutions.

Table 3
Future research directions proposed by seminal reviews on HLPs since 2008.

Reference	Future Research Directions
Alumur and Kara (2008)	Gaps in specific combinations of formulation components, Use of multiple, possibly conflicting objectives, Development of more realistic studies addressing real-life situations
Kara and Taner (2011)	Consideration of competition, Incorporation of a non-linear cost function
Campbell and O'Kelly (2012)	Use of more realistic transportation costs and service measures, Interdisciplinary aspects and different habits in research communities, Presentation and documentation of results
Farahani et al. (2013)	Downstream traffic generation (hubbing effect), Skewed flows towards non-hub connections, Plane-based formulations, Multi-criteria decision making, Reliability, sustainability, global logistics applications, Uncertainty and scalability of methods
Essaadi et al. (2016)	Novel research themes along the grid's dimensions
Contreras and O'Kelly (2020)	Models considering competitive environment, collaborations, mergers, and acquisitions, Optimal evolution of hub networks and choice of transportation mode
Alumur et al. (2021)	Better modeling of economies of scale, Incorporation of time, More sophisticated objectives/multiple criteria, Realistic models reflecting industry changes or travel patterns, Integration of HLP with other problems, Use of more realistic data, Transforming results to insights
Khaleghi and Eydi (2022)	Multi-objective models, Sustainability aspects, Focus on continuous-time models, Incorporation of uncertainty/waiting times at hubs, Competitive/pricing models
Wandelt et al. (2022)	Choosing appropriate solution methods, Predicting and explaining extreme computation times, Focus on scalability, Use of data mining and artificial intelligence
Sharma et al. (2025)	Integration of machine learning and dynamic factors, Trade-off between optimal solutions and heuristics, Sustainable hub location models, Collaborative/coopetitive hub networks, Impact of emerging technologies in location decision making

4.2. Insight-based research questions and presentation

Various existing reviews have highlighted the need to focus more on insights than results (Alumur et al., 2021; Campbell & O'Kelly, 2012). Particularly, Campbell and O'Kelly (2012) recalled the following paraphrase: *The purpose of hub location research is insight, not numbers*. In other words, *The goal of hub location modeling is to develop a better understanding of the nature of optimal (or near-optimal) solutions* (Alumur et al., 2021). Here, the identification of metrics, which are independent of problem instances (e.g., built from specific ratios), is very beneficial for researchers and also challenging area worth of research, because it allows to cross-check results for being reasonable. Below we discuss this challenge in greater detail.

Papers should present at least some empirical data on actual objective function solutions obtained (instead of just run times). The reasons are that they allow further research runs to have a database of existing best-known solutions for comparison purposes. When it may prove impossible to solve a problem to optimality it is useful to have a lower bound. Further, detailed examination of runs that provide trouble for an algorithm may provide insights into the spatial arrangements issues that are particularly problematic. Identification of so-called hard cases is exceptionally useful. Further, since the algorithm can produce both location and allocation information (and these are not directly discerned from just knowing hub locations) there is some benefit of a compact way to portray the allocation part of the solutions. Researchers may also benefit from the careful numerical investigation of actual solutions and their visualization/discussion in a paper. Insights from attempts to tackle a specific problem may provide information for researchers with other problem domains. It is also perhaps worth retaining a focus on the real underlying problem and keeping an eye on common sense expectations. Thus, as the "alpha" discount value becomes lower, a problem with fixed facility costs will tend to want to open more facilities. If a problem or set of results produces counter intuitive results it may be worth checking for errors. It can also be a useful check to run the model with special case parameters just to make sure that the results collapse correctly (for example if alpha = 0, we can be sure that the upper and lower bound are identical and the results are the same as the p-median problem).

4.3. Sustainability

Incorporating sustainability into HLP research is essential for addressing the environmental challenges of the 21st century (Farahani et al., 2013; Khaleghi & Eydi, 2022; Sharma et al., 2025). The environmental impact of aviation – which has long been neglected by the industry in favor of growth – has become a major concern (Karakoc et al., 2022). As the demand for more sustainable air transportation grows, there is a pressing need for models that prioritize environmental sustainability. This involves optimizing hub locations to minimize greenhouse gas emissions, reduce energy consumption, and promote the use of renewable resources. Sustainable HLPs also consider the long-term environmental impact of hub operations, including waste management and resource conservation. By integrating sustainability goals into the decision-making process, researchers can develop solutions that balance economic efficiency with environmental responsibility, contributing to a greener and more resilient mobility as well as supply chain.

Integrating sustainability into hub location problems presents significant modeling challenges that require a comprehensive approach. Traditional hub location models focus on optimizing costs or efficiency, often using linear or mixed-integer programming techniques. However, incorporating sustainability introduces additional complexities that necessitate the integration of environmental and social metrics into these models. One of the primary modeling challenges is the definition and quantification of sustainability metrics. Metrics such as carbon emissions, energy consumption, and waste generation must be accurately measured and incorporated into the model. This requires the development of adequate mathematical formulations that can capture these metrics without compromising the model's solvability. For instance, the carbon footprint of a hub location can be modeled as a function of distance and emission factors: Emissions $= \sum_{i,j} d_{ij} \cdot e_{ij} \cdot x_{ij}$, where d_{ij} is the distance between locations i and j, e_{ij} is the emission factor, and x_{ij} is a decision variable indicating the flow between locations. Another significant challenge is the management of trade-offs between efficiency, economic and sustainability objectives. These trade-offs often require the use of multi-objective optimization techniques to find Pareto-optimal solutions that balance competing goals. For example, minimizing transportation costs may conflict with reducing carbon emissions, necessitating a compromise solution that satisfies both objectives to a certain degree. Moreover, factors such as fluctuations in energy prices, changes in environmental regulations, and unpredictable weather patterns introduce stochastic elements into the model. Robust optimization and scenario-based approaches are essential for addressing these uncertainties and ensuring the model's resilience. In conclusion, addressing the modeling challenges of integrating sustainability into hub location problems requires advanced mathematical techniques, robust data management, and a coordinated approach that considers economic, environmental, and social factors.

4.4. Robustness

The consideration of robustness in hub location models is becoming increasingly important, particularly in addressing disruptions such as facility/equipment failures or intentional attacks/intrusion attempts (Farahani et al., 2013; O'Kelly, 2025). Researchers have developed robust optimization methods to enhance the adaptability of hub networks, ensuring network continuity despite adverse events. Delay is considered a disruption that must be mitigated. Robust models account for various risk factors to maintain high service levels under challenging conditions. This involves identifying backup hub locations, diversifying passenger or supply routes, and implementing redundancy measures, as seen in models that account for link failures. Enhancing the robustness of hub networks is crucial for building reliable and adaptable transport systems that can respond effectively to unexpected challenges.

Integrating robustness into hub location problems using solvers like CPLEX involves specific mathematical formulations and techniques to handle disruptions. Robust optimization aims to find solutions that are feasible across a range of scenarios, including worst-case situations. In CPLEX, this can be formulated using a two-stage approach. The first stage involves determining the location and capacity of hubs, while the second stage adjusts operational decisions based on realized disruptions. Mathematically, this can be expressed as $\min\left\{c^T x + \max_{s\in S} Q(x,\xi_s)\right\}$, where x represents the first-stage decision variables (e.g., hub locations), c is the cost vector, S is the set of scenarios, and $Q(x,\xi_s)$ is the optimal value of the second-stage problem under scenario s with realization ξ_s. This deterministic approach ensures that the system can withstand the worst-case scenario, promoting reliability and resilience. Dynamic programming can be used to evaluate recovery strategies over time. Simulation models, often implemented in specialized software, can assess the impact of disruptions and test recovery protocols. These models simulate various disruption scenarios and evaluate the system's ability to return to normal operations. By leveraging these formulations and techniques in CPLEX and other solvers, decision-makers can develop hub location models that are not only efficient but also robust and resilient against disruptions, ensuring reliable performance under challenging conditions.

4.5. Uncertainty

In a related context to robustness, addressing uncertainty is another challenge in HLP research (Farahani et al., 2013; Khaleghi & Eydi, 2022). Air transport as well as modern supply chains are subject to various uncertainties, including demand fluctuations, supply disruptions, and regulatory changes. Incorporating uncertainty into HLP models allows for more realistic and adaptable solutions. Stochastic optimization techniques can be employed to model uncertain parameters, such as demand, cost, and travel times. By considering multiple scenarios and their probabilities, researchers can develop robust strategies that minimize risks and optimize performance under uncertainty.

Incorporating uncertainty into hub location problems presents various technical challenges. Traditional hub location models assume deterministic parameters, such as fixed costs and demand patterns. However, real-world scenarios are fraught with uncertainties, including fluctuating demand, variable transportation costs, and disruptions in supply chains. These uncertainties can stem from various sources, such as economic fluctuations, natural disasters, or changes in consumer behavior. Integrating these uncertainties into the model requires sophisticated probabilistic and stochastic approaches. One of the primary challenges is the computational complexity involved in solving stochastic models. These models often require simulating a wide range of scenarios to capture the potential variability in inputs. This necessitates advanced algorithms and substantial computational resources, which can be prohibitive for many organizations. Moreover, the quality and availability of data significantly impact the accuracy of stochastic models. Historical data may not always be indicative of future trends, especially in rapidly changing environments. Therefore, obtaining reliable and up-to-date data is crucial for developing robust models that can withstand real-world variability. Another challenge is the need for interdisciplinary expertise. Effective integration of uncertainty requires knowledge in operations research, statistics, and data science. Collaboration among experts in these fields is essential to develop comprehensive models that account for various types of uncertainties. Furthermore, organizations must be willing to adopt a more flexible and adaptive approach to decision-making. This involves embracing uncertainty as an inherent part of the planning process and being prepared to adjust strategies as new information becomes available. To overcome these challenges, it is important to invest in advanced analytics tools and technologies. Machine learning and artificial intelligence can play a pivotal role in enhancing the predictive capabilities of stochastic models. These technologies can help identify patterns and trends in data that may not be immediately apparent through traditional methods. In conclusion, while incorporating uncertainty into hub location problems is technically challenging, it is a necessary step towards navigating the complexities of an uncertain world.

4.6. Parallelization

Solving HLPs is an inherent challenge, given their NP hardness, be it for obtaining optimal solutions at medium scale or heuristic solutions at large scale (Farahani et al., 2013; Sun et al., 2017; Wandelt et al., 2022). Accordingly, parallelization techniques have the potential for significantly enhancing the computational efficiency of HLP research, allowing to explore multiple solution spaces simultaneously. In this context, GPU and quantum computing have enormous potential to speed up solution techniques by accelerating computational processes. GPUs can massively parallelize operations with thousands of simple, parallel processing units. Quantum computing, on the other hand, with its ability to solve complex optimization problems more efficiently than classical computers, may revolutionize the approach to HLPs by identifying optimal solutions from a combinatorial number of possibilities. The early pursuit of this advanced computing technologies – and deeper appreciation of these computer science-related optimizations by the operations research community – will enable significant computational advances, better solutions for large scale instances, and possibly more sophisticated modeling techniques. We discuss a few parallelization-related specific concerns below.

Exact methods, while allowing for optimal solutions, often face computational challenges that parallelization can mitigate. Integer Programming is inherently sequential due to its reliance on solving linear systems, but certain aspects, such as constraint evaluation and feasibility checks, can often be parallelized. Benders Decomposition, which breaks down the problem into a master problem and subproblems, is naturally suited for parallelization. The subproblems can be solved simultaneously on different processors, making it efficient for large-scale problems with decomposable structures. B&B can benefit from parallelization by exploring multiple branches of the search tree concurrently, though synchronization is required to manage partial and global best solutions. B&C extends this by incorporating cutting planes, which can also be computed in parallel, though the overall process remains computationally intense. Lagrangian Relaxation can parallelize the evaluation of the Lagrangian dual across different subproblems, reducing complexity but still providing lower-bound solutions. Heuristic methods are, in general, more amenable to parallelization due to their iterative and stochastic nature. For GAs, one can parallelize the evaluation of fitness functions across a population of solutions, significantly speeding up the search process. Tabu Search might explore multiple neighborhoods simultaneously, though careful management is needed to avoid redundancy and ensure effective search space coverage. GRASP combines greedy randomization with local search, allowing parallel execution of both phases to accelerate convergence, though solution quality still depends on controlling randomness. VNS can parallelize the exploration of different neighborhood structures, balancing exploration and exploitation effectively. In summary, while exact methods may benefit from parallelization in specific components, heuristic methods are likely to leverage parallelization more comprehensively for large-scale and complex hub location problems where approximate solutions are acceptable. At the same, researchers – especially from outside the HLP community – should be cautious: Some problems which turn out the be parallelized nicely in a rather straightforward way, might be solvable well by existing algorithms either. For example, Berrajaa et al. (2023) deploy a GPU based genetic algorithm for the multiple allocation p-hub problem. While this is notable, it is appropriate to point out that this version of the problem is well-handled by exact methods. Perhaps more challenging would be an attack on the single allocation problem, which seems to have promising connections to quadratic assignment tasks, and these have been approached with novel architecture such as quantum computing. Similar problems exist when comparing against the wrong baseline heuristic. For instance, VNS-based methods have shown satisfying results for various HLPs; and they should be the ultimate baseline for comparison of these modern, highly parallelizable algorithms / implementations.

4.7. Temporal aspects

Incorporating temporal aspects into HLP research is vital for capturing the dynamic nature of logistics and transportation networks (Khaleghi & Eydi, 2022). Temporal models consider time-dependent factors, such as peak demand periods, seasonal variations, and real-time traffic conditions. Particularly, multi-period hub location problems could be further generalized by incorporating temporal aspects at various and variable scales, accounting for seasonal variations and peak demand periods, allowing models to adapt dynamically to fluctuating market conditions. By integrating real-time data and predictive analytics, these models can optimize hub operations and routing decisions based on current/future traffic and demand patterns, enhancing responsiveness and efficiency. Additionally, exploring the long-term impacts of temporal factors, such as infrastructure developments or regulatory changes, can provide insights into the evolving requirements of hub networks over time. Finally, open data regarding temporal aspects of HLPs is needed by the community.

Integrating temporal aspects into hub location problems, particularly in the context of multi-period hub location problems, introduces a range of modeling challenges that require sophisticated approaches to address effectively. Multi-period hub location problems involve determining the optimal placement of hubs over multiple time periods, considering dynamic changes in demand, capacity, and other temporal factors. This temporal dimension adds layers of complexity to an already intricate problem, necessitating advanced modeling techniques. One primary challenge is capturing dynamic demand patterns over time, which requires time-dependent variables and robust forecasting techniques to accurately predict future demand scenarios. Capacity planning decisions must also be modeled, involving the adjustment of hub capacities over time to accommodate changing demand, which can be capital-intensive and require long-term planning. Time-dependent costs, such as transportation and labor, must be integrated into the model using financial forecasting techniques and discounted cash flow analysis. Uncertainty and risk management are crucial, as temporal aspects introduce additional uncertainties and risks, necessitating the use of stochastic programming and scenario analysis to develop robust solutions. Inter-temporal trade-offs between short-term and long-term objectives must be balanced using multi-objective optimization techniques. Additionally, future technological advancements that could impact hub operations must be considered, incorporating technological forecasting and scenario planning into the model. Finally, researchers could focus on more continuous time models, where decisions can be made at any point in time in the planning horizon, contrary to discrete-time planning. Addressing these modeling challenges requires a comprehensive and dynamic approach, leveraging advanced optimization techniques, robust forecasting methods, and scenario analysis to develop multi-period hub location solutions that balance current and future needs, manage uncertainties, and adapt to changing conditions, thereby enhancing operational efficiency and supporting long-term strategic planning and sustainability goals.

4.8. Interdisciplinary aspects

An interdisciplinary approach is crucial for tackling the multifaceted challenges of HLPs. By combining insights from operations research, urban planning, environmental science, economics, as well as computer science, researchers can develop holistic solutions that address the complexities of modern supply chains. Traditionally, hub location problems have been approached mostly from an operational research standpoint, emphasizing mathematical modeling and optimization algorithms. Cooperation with other disciplines possibly fosters a holistic approach to hub location research. For instance, a collaborative effort could involve urban planners identifying potential sites, economists evaluating their financial viability, environmental scientists assessing their ecological impact, and sociologists ensuring that the chosen locations serve the needs of diverse communities. This interdisciplinary cooperation can lead to more comprehensive and sustainable solutions that consider a broader range of factors beyond mere operational efficiency.

Existing research is often siloed based on publication venues and domain-specific research groups, who have significant potential to establish gated communities which generate difficulties for cross-disciplinary perspectives. A striking example is that many operations research researchers still do not consider scalability as a major challenge for their models, often undervaluing the need for the development of highly efficient heuristics. A more collaborative approach enhances the applicability of HLP models and solutions, driving future progress in sustainable and efficient logistics networks. Moreover, valuation among researchers from different domains can enhance the relevance and applicability of hub location research. By recognizing and incorporating diverse perspectives, researchers can develop models and frameworks that are more robust and adaptable to real-world complexities. This valuation also encourages knowledge sharing and innovation, as researchers from different fields bring unique methodologies and insights to the table. Finally, competitive airline strategies and market dynamics are important; for instance, a merger between two airlines that operate hubs would not necessarily alter the hub location problem but could lead to the discontinuation of a hub. Ultimately, an interdisciplinary approach to hub location research can lead to more informed decision-making, improved resource allocation, and enhanced community outcomes, benefiting both academic research and practical applications.

4.9. Versatile hubs

Existing studies usually focus on passenger or cargo applications. Designing versatile, modern hubs is essential for addressing the diverse and evolving needs of transportation and supply chains. Versatile hubs can handle multiple types of cargo, accommodate various transportation modes, and adapt to changing operational requirements, while potentially reducing physical dependencies. This involves integrating flexible and digital infrastructure, to support a wide range of activities. Versatile hubs also incorporate advanced technologies, such as automation and robotics, to enhance operational efficiency and responsiveness.

By developing models for the incorporation of versatile hub solutions, researchers can create adaptable and resilient logistics networks that meet the dynamic demands of the market. We discuss this direction in greater detail below.

One major aspect of versatility is being virtual. In air transport, virtual hubs can facilitate the efficient use of aircraft capacity by matching supply with demand through digital marketplaces. Airlines can collaborate more effectively, sharing capacity and routes to maximize load factors and reduce empty flights. This not only enhances operational efficiency but also contributes to sustainability goals by minimizing fuel consumption and emissions. For logistics, virtual hubs offer a decentralized model that can streamline the movement of goods. By leveraging data from IoT devices and real-time tracking systems, logistics providers can dynamically reroute shipments, avoiding congestion and delays. Furthermore, virtual hubs can integrate with advanced analytics and machine learning algorithms to predict demand patterns and optimize routing, leading to improved service levels and reduced costs.

4.10. Emerging modes and their integration into hub location models

The integration of emerging transportation modes, such as drones and autonomous vehicles, as well as established ones, such as High-Speed Rail (HSR), presents new opportunities and challenges for HLP research (Contreras & O'Kelly, 2020; Sharma et al., 2025). These modes offer the potential to enhance the efficiency and sustainability of airport hub operations, and – at the same time – may act as a major threat. In China, for instance, the progressive dissemination of high-speed rail, in urban and rural areas, has led to a significant reduction of markets: City pairs which were solely solved by air transport in the past, may now even see no air connectivity at all. A striking example is the connection between Beijing and Shanghai, at a distance of approximately 1,200 km. The incorporation of modern competitors require the development of new models and infrastructure to support their effective integration into existing air transportation networks. Researchers must address the unique operational and regulatory challenges associated with emerging modes, such as airspace management for drones and safety standards for autonomous vehicles. We discuss a few related specific concerns below.

Integrating hub location problems with emerging transportation modes, such as HSR and air taxis, presents unique technical and mathematical modeling challenges. Traditional hub location models are designed for conventional transportation networks, focusing on minimizing costs and maximizing efficiency through centralized hubs. However, emerging modes like HSR and drones introduce complexities due to their distinct operational characteristics and infrastructure requirements. High-Speed Rail (HSR) systems, characterized by their speed and capacity, require hubs that can handle high passenger volumes and frequent services. Modeling challenges include accommodating the increased flow and ensuring seamless connectivity with existing transportation networks. The placement of HSR hubs must consider not only geographical factors but also the economic and social impacts on surrounding regions. Additionally, the successful integration of HSR necessitates multi-modal hubs that can efficiently transfer passengers and cargo, adding layers of complexity to the optimization problem. Air taxis, as an emerging urban air mobility solution, present further challenges due to their vertical takeoff and landing capabilities and the need for dedicated landing sites. The optimal placement of air taxi hubs must account for urban density, noise restrictions, and safety regulations. Moreover, air taxis operate in a three-dimensional space, requiring advanced spatial modeling techniques to determine efficient flight paths and hub locations. Integrating air taxis with existing air and ground transportation networks involves creating intermodal hubs that facilitate smooth transitions between air and ground travel, posing additional logistical and infrastructural challenges. Mathematically, these integrations demand sophisticated models that can handle multi-objective optimization, considering factors like cost, travel time, environmental impact, and passenger convenience. Stochastic modeling is essential to account for uncertainties in demand and operational disruptions. Finally, the formulation of competitive and cooperative aspects adds another layer of game theoretic complexity, which requires efficient solution techniques. Addressing these challenges requires interdisciplinary collaboration, leveraging advancements in operations research, transportation engineering, and urban planning to develop robust and adaptable hub location models for the future of transportation.

5. Conclusion

We have reviewed the state of the art in hub location problems in this study. Unlike previous surveys, which often focus on specific aspects or subsets of the problem, our approach is distinctive in its holistic integration of insights from multiple surveys, critical analysis of their contributions and propositions for future work. The field of hub location research continues to evolve, incorporating robustness, sustainability, and new transportation technologies. These expansions ensure that hub networks remain efficient, resilient, and adaptable to the demands of modern logistics and transportation systems. We believe that the challenges for DISRUPTIVE research regarding hub location problems helps the community in terms of coordinated efforts for reaching these expansions.

CRediT authorship contribution statement

Morton O'Kelly: Writing – review & editing, Writing – original draft. **Xiaoqian Sun:** Writing – review & editing, Writing – original draft. **Sebastian Wandelt:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The authors report financial support was provided by National Natural Science Foundation of China. They authors have no other known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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